

Neural Estimation of Asset–Factor Exposure Matrices under Multiple Candidate Factor Structures for Clustering

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Abstract

Keywords:

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1 Introduction

The discovery of latent structure in high-dimensional time series has long been central to understanding complex systems such as financial markets, macroeconomic indicators, and cross-asset dependencies. In these settings, lead-lag relationships, the temporal dependencies that indicate movements in one variable precede those in another, play a crucial role in forecasting, clustering, and identifying causal or informational hierarchies. A standard modeling paradigm for such systems is the multi-factor model, in which each asset’s return is expressed as a linear combination of latent or observable factors with distinct exposures and noise components. Estimating these exposures and uncovering the temporal relationships among them remain fundamental challenges, particularly in the presence of noise and overlapping factor influences.

Recent advances have sought to address these issues through clustering-driven approaches that jointly exploit factor structure and temporal alignment. In particular, Zhang et al. (2025) proposed a robust pipeline for detecting lead-lag relationships in lagged multi-factor models via applying clustering algorithms to an enlarged universe of subsequences, and discovering their directional dependencies through intra-cluster lag. This line of work represents a significant step forward in the detection of high-dimensional lead-lag structure. However, reliance on subsequence extraction, homogeneous factor loadings, and static clustering limits the interpretability in factor space, and restricts the model’s ability to learn representations directly from the data.

In this paper, we extend this literature by introducing a neural estimation framework that integrates factor learning, clustering, and lead-lag inference within a unified architecture. Our approach models the asset-factor relationship through a signed exposure matrix estimated by a neural network consistent with the following observation model:

$$R = \hat{W}\hat{H}^\top + E,$$

where $\hat{W}$ captures the learned exposures and $\hat{H}$ represents a panel of candidate factors. After top- k sparsification and row normalization, the exposures define a cosine-equivalent geometry in which we perform density-based clustering (DBSCAN) to obtain coherent asset groupings. Cluster-level series are then constructed and analyzed through lagged cross-correlation to infer directed lead-lag relationships amongst the clusters.

Our main contributions are as follows:

1. We propose a neural estimation framework to learn signed factor exposures under multiple candidate factor structures, unlike prior methods which discover only single factor structures.
2. We introduce a geometry-preserving embedding of assets that enables robust and interpretable clustering in exposure space.
3. We develop a cluster-level lead-lag inference procedure yielding a directed Γ matrix summarizing inter-cluster temporal dependencies.
4. We validate the method on synthetic and financial data, demonstrating improved robustness and interpretability relative to existing clustering-based approaches.

The remainder of the paper is organized as follows. Section 2.1 presents the model setup, Section 2.2 details the neural estimation framework, and Section 2.3 describes the clustering and lead-lag inference procedures. Section 3 introduces the data generation process, Section ?? reports experimental results, and Section 5 concludes.

2 Methods

2.1 Model Setup

We consider n assets, k factors, and T time steps. Let $R \in \mathbb{R}^{n \times T}$ denote the asset return series, $\hat{H} \in \mathbb{R}^{T \times k}$ a candidate (or estimated) factor return series, and $\hat{W} \in \mathbb{R}^{n \times k}$ the *signed* factor exposure matrix. The observation model is

$$R = \hat{W} \hat{H}^\top + E, \quad (1)$$

with residual term E . We emphasize the importance of a signed exposure matrix, as it provides intuition for inverse relationships between assets.

2.2 Neural Estimation Framework

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2.3 Clustering

We cluster assets in the space of signed exposures. Throughout, rows of $\hat{W}$ index assets and columns index factors.

Construction of the embedding. From $\hat{W}$ we form $X \in \mathbb{R}^{n \times k}$ by the two-step transformation

$$(i) \text{ top-}k \text{ per row: } X_{i:} \leftarrow \mathcal{T}_{k_{\text{top}}}(\hat{W}_{i:}), \quad (ii) \text{ row normalization: } X_{i:} \leftarrow \frac{X_{i:}}{\|X_{i:}\|_2}, \quad (2)$$

where $\mathcal{T}_k$ keeps the k entries of largest magnitude in a vector and zeros the remainder. The normalization renders Euclidean distances on X equivalent to cosine dissimilarities on the signed directions:

$$\left\| \frac{x}{\|x\|_2} - \frac{y}{\|y\|_2} \right\|_2^2 = 2(1 - \cos \theta(x, y)).$$

We choose to analyze cosine geometry, as it is unaffected by magnitude of exposure vectors; therefore, neglecting volatility and focusing solely on direction. Variants (e.g., IDF reweighting, positive/negative duplex channels) can be analyzed analogously and are deferred to the appendix.

Density-based clustering (DBSCAN). Given the embedding (X, d) and parameters $(\varepsilon > 0, m \in \mathbb{N})$, define the ε -neighborhood $N_\varepsilon(x) = \{y \in X : d(x, y) \leq \varepsilon\}$. A point x is a *core* point if $|N_\varepsilon(x)| \geq m$. A point y is *directly density-reachable* from a core point x if $y \in N_\varepsilon(x)$; *density-reachable* is the transitive closure of this relation. Two points are *density-connected* and can be clustered if there exists a core point from which both are density-reachable. A cluster is a maximal density-connected set; points that are not density-reachable from any core are labeled *noise*, and points with $|N_\varepsilon(x)| < m$ that lie in the neighborhood of a core point are *border* points.

Here, Euclidean threshold ε corresponds to a cosine similarity lower bound $\cos \theta \geq 1 - \varepsilon^2/2$. DBSCAN is appropriate here because it does not require pre-specifying the number of clusters, is robust to isolated points, and can recover non-spherical clusters in exposure space.

Choice of neighborhood radius. Let $d_{(1)} \geq \dots \geq d_{(n)}$ be the distances ordered m -th nearest-neighbor for the rows of X . We select $\hat{\varepsilon}$ by the elbow of this curve, defined as

$$i^* \in \arg \max_{1 \leq i \leq n} \text{dist}((i, d_{(i)}), \ell(i)),$$

where ℓ is the chord joining $(1, d_{(1)})$ and $(n, d_{(n)})$, and set $\hat{\varepsilon} = d_{(i^*)}$ (optionally scaled by a constant factor $\alpha \in (0, 1]$). To protect against degenerate results, we adopt the following continuation scheme: while the number of discovered clusters (excluding noise) is below a target $C_{\min}$ and an iteration budget remains, we replace $\varepsilon \leftarrow \gamma \varepsilon$ with $\gamma > 1$ and refit.

Cluster representatives and series. Let $\mathcal{C}_1, \dots, \mathcal{C}_C$ denote the clusters. We define representative exposures.

$$\bar{w}_c = \frac{1}{|\mathcal{C}_c|} \sum_{i \in \mathcal{C}_c} \hat{W}_{i:} \in \mathbb{R}^k, \quad c = 1, \dots, C,$$

and the corresponding cluster-level series by projection onto the factors,

$$s_c(t) = \bar{w}_c^\top \hat{h}_t, \quad S = \bar{W} \hat{H}^\top \in \mathbb{R}^{C \times T}, \quad (3)$$

where $\bar{W}$ stacks the $\bar{w}_c$ as rows.

Inter-cluster lead-lag (as implemented). For clusters $i \neq j$, set $x = s_i$ and $y = s_j$, and define the centered series $x_z = x - \bar{x}$, $y_z = y - \bar{y}$ with the fixed normalizer

$$D = \|x_z\|_2 \|y_z\|_2 + 10^{-12}.$$

For integer $L \in [-L_{\max}, L_{\max}]$, the normalized cross-correlation uses overlapped slices and this global normalizer,

$$\text{NCC}_{ij}(L) = \begin{cases} \frac{\sum_{t=1}^{T-L} x_z(t) y_z(t+L)}{D}, & L \geq 0, \\ \frac{\sum_{t=1-L}^T x_z(t) y_z(t+L)}{D}, & L < 0. \end{cases}$$

We select

$$L_{ij} \in \arg \max_{L \in [-L_{\max}, L_{\max}]} |\text{NCC}_{ij}(L)|, \quad \rho_{ij} = \text{NCC}_{ij}(L_{ij}),$$

with the convention $L_{ij} > 0$ meaning that i leads j by $|L_{ij}|$ time steps and $L_{ii} = 0$.

Gamma matrix. The inter-cluster lag matrix $\Gamma \in \mathbb{Z}^{C \times C}$ collects the best lags,

$$\Gamma_{ij} = L_{ij}, \quad \Gamma_{ii} = 0,$$

and the matrix $\rho \in \mathbb{R}^{C \times C}$ stores the corresponding signed scores $\rho_{ij} = \text{NCC}_{ij}(\Gamma_{ij})$. For graphical summaries, we consider the directed edge $i \rightarrow j$ when $\Gamma_{ij} > 0$ and $|\rho_{ij}| \geq \tau$.

3 Data Generation

4 Experiments

5 Conclusions

References

- Y. Zhang, M. Cucuringu, A. Y. Shestopaloff, and S. Zohren. Robust Detection of Lead–Lag Relationships in Lagged Multi-Factor Models. *Applied Mathematical Finance*, pages 1–37, 2025. doi:10.1080/1350486X.2025.2544272.